Mamba: Linear-Time Sequence Modeling with Selective State Spaces
Albert Gu, Tri Dao

Abstract
Foundation models based on the Transformer architecture have achieved immense success, but suffer from computational inefficiency on long sequences due to quadratic attention complexity. Structured State Space Models (SSMs) offer linear-time scaling with sequence length, but struggle with content-based reasoning. We introduce Mamba, a selective state space architecture that makes SSM parameters input-dependent, allowing the model to selectively propagate or forget information along the sequence dimension. Mamba achieves linear-time training and inference complexity, processing sequences up to 1 million tokens efficiently on modern GPUs. Mamba-3B matches or exceeds Transformers of similar size across language modeling and downstream evaluations, while offering 5x higher inference throughput.

1. Introduction
Transformers require O(N^2) time and memory for sequence length N. Prior sub-quadratic alternatives like linear attention and traditional SSMs lacked selectivity?the ability to filter irrelevant context. Mamba integrates selection mechanisms directly into the continuous-time state space matrices (Delta, B, C) as a function of the input token.

2. Hardware-Aware Hardware Algorithm
To overcome memory bandwidth bottlenecks on GPUs, Mamba introduces a hardware-aware parallel scan that computes the recurrent state in SRAM rather than transferring intermediate states through HBM. This achieves a 3x speedup over standard scan implementations.

3. Findings
On the Pile benchmark, Mamba-2.8B achieved a perplexity of 5.89, outperforming Pythia-2.8B (perplexity 6.22) and LLaMA-3B equivalents. At sequence length 8192, Mamba achieved 5.2x faster inference generation than equivalent Transformer models. 4. Limitations
State space models exhibit lower in-context learning recall on needle-in-a-haystack tasks with more than 100 distracting entities compared to full softmax attention.